

Ques. WHY IS GAIN AND INPUT IMPEDANCE OF AN IDEAL OP-AMP INFINITY?

Akash Belakwadi

Ans. An ideal operational amplifier (op-amp) has infinite input resistance, zero output resistance, infinite open-loop gain and infinite bandwidth. However, a practical op-amp has performance limitations.

An operational amplifier's input voltage V_o is the voltage difference between the two input terminals. Let us say R_i represents input resistance and R_o the output resistance. Input voltage is amplified by open-loop gain, or A_{OL} (op-amp voltage gain without external negative feedback), to produce the output (V_o). Ideally, op-amp amplifies only the difference in voltage between the two, which is called differential input voltage. So,

$$V_o = A_{OL} V_D$$

As with other voltage amplifiers, op-amp voltage gain is output voltage divided by input voltage. Direct output voltage (V_o) can be measured as the voltage difference between the output terminal and ground level, while input voltage is the difference between the voltage levels at the two input terminals, that is, differential input voltage (V_D).

To understand why an op-amp has infinite open-loop gain, also defined as open-loop voltage gain (A_{OL}) or large signal voltage gain, let us for instance refer LM741 data sheet which specifies gain of 50V/mV minimum and 200V/mV typical. This means that a 1mV difference between the two input terminals would produce a minimum of 50V output. Thus (differential) input signal is amplified by a factor of 50,000.

The typical amplification is 200,000, and maximum amplification can be considerably greater. This is

similar to the situation with h_{FE} of a BJT, where the manufacturer specifies minimum, typical and maximum values. Such voltage gains are too high for many applications, but are very suitable as open-loop gain for circuits using negative feedback.

An op-amp has infinite input impedance to avoid loading effect, so that it does not draw any current from a signal source and does not cause any voltage drop across a source resistance, so that all of the input signals appear across op-amp input terminals.

Q2. CAN YOU PLEASE HELP ME DESIGN A SONAR WATER-LEVEL METER AND PROVIDE LINKS FOR FURTHER INFORMATION?

Rajatsekhar

A2. There are several methods of designing a sonar water-level meter.

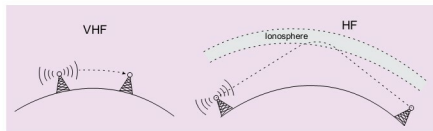


Fig. 1: HF and VHF signal propagation

An ultrasonic water level sensor uses ultrasonic distance-measuring technology for accurate non-contact water-level monitoring. One such project was published in October 2015 issue of *Electronics For You*. Please find below the link for your reference:

<http://electronicsforu.com/electronics-projects/hardware-diy/microcontroller-sonar-water-level-meter>

In this sonar water-level meter project, fabrication of a sonar contact-less, wireless water-level indicator is

described. It is capable of measuring water levels by using ultrasonic sensors. Ultrasonic sensor HC-SR04 is used to measure the distance.

Q3. WHY IS HF RADIO USED FOR COMMUNICATIONS OVER LONG DISTANCES, INSTEAD OF VHF?

Joy Mukherjee

A3. High frequency (HF) radio is used for communications over long distances, mostly in remote regions of the developed world. This happens mainly because HF radio requires minimal infrastructure and can also be used to communicate with existing very-high-frequency (VHF) radio systems.

HF propagates through ionospheric refraction, so long-path communication is possible through multi-hop propagation at various frequencies.

On the other hand, VHF waves travel in line-of-sight, thus curvature of Earth between transmitting and receiving station becomes a hindrance. Thus, long-distance VHF transmission is not possible without the use of repeaters (Fig. 1).

Answers compiled by EFY senior application engineer, Nidhi Kathuria. Letters and questions for publication may be addressed to Editor, *Electronics For You*, D-97/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (e-mail: editsec@efy.in) and should include name and address of the sender